Presentation Script: Structural Loading on Appendages

Slide 1: Title

Welcome everyone. Today, we'll explore **Structural Loading on Appendages**, focusing on how wings and fins in nature manage aerodynamic forces. We'll discuss the physics behind it and walk through a practical example using the albatross takeoff scenario.

Slide 2: Motivation

Why does this matter?

- Nature offers elegant solutions for flight.
- Understanding these mechanisms helps us design efficient and safe aircraft.
- Structural loading is key to performance and avoiding failure.

Slide 3: Propulsive Appendages in Nature

Let's look at how animals use their appendages:

- Fish use fins to steer and push through water.
- Birds and insects use wings to generate lift and control direction.
- These appendages must endure loads without breaking or deforming uncontrollably.

Slide 4: Gliding Ray and Lift Distribution

This figure shows a gliding ray:

- It uses its fins to glide, generating lift like a bird's wing.
- The figure on the right simplifies this concept using a twisted wing.
- This introduces the idea of lift distribution across a span.

Slide 5: 3D Effects – Root and Tip Vortices

- Real wings are not infinite.

- At the root and tip, 3D effects become important.
- Vortices form at the tips and alter lift distribution.
- Prandtl's lifting-line theory helps model this behavior mathematically.

Slide 6: Bio-Inspired Propulsion

This slide shows wings of a samara and a bio-inspired propeller blade:

- Both are broken into 2D slices for analysis.
- The inflow velocity and rotation define how much force each slice feels.

Slide 7: Modeling Structural Forces

We simplify by analyzing cross sections:

- Forces include lift (C_L), drag (C_D), normal (C_N), and axial (C_X).
- Moments are bending in two directions and twisting (torsion).

Slide 8: Force Components on a Section

This airfoil diagram helps explain the forces:

- Lift acts upward, drag backward.
- From these we compute the normal and axial components that matter for structure.

Slide 9: Albatross Takeoff Scenario

Let's move into a real-world case:

- The albatross takes off at a 30° angle with 15 m/s speed.
- Its wingspan and geometry are known.
- We analyze the forces and structural moments involved at the moment of lift-off.

Slide 10: Takeoff Modeling Example

We model the wing using:

- An elliptical planform
- Given angle of attack and velocity
- Using simple aerodynamic formulas, we compute:
  - Lift and drag coefficients
  - Resulting force components
  - Flapwise, edgewise, and torsional moments

Conclusion: These simplified models give us insight into how wings bear loads, showing the balance between aerodynamic efficiency and structural integrity.